INOS/SSOT/HEMPIN

Intelligent Network for Open Sourced Single Source of Truth for Hemp Industry

EIC Pathfinder Open 2026 — Part B Technical Description

#@APP-FORM-HEEICPAOP@#

1. Excellence

1.1 Long-term vision

The Vision: Verified Truth Infrastructure for Planetary Regeneration (2045)

By 2045, global supply chains will operate on a layer of verified truth infrastructure that enables efficient cross-border trade while automatically enforcing regulatory compliance and funding environmental regeneration. This infrastructure will not be owned by any single corporation or state, but will function as a sovereign, open protocol — analogous to TCP/IP for physical goods. The INOS/SSOT/HEMPIN project is the foundational proof-of-principle research phase toward this vision, using the hemp industry as its testbed domain.

The Radically New Technology: Regulatory-Encoding Knowledge Graphs

Current supply chain technologies (blockchain, ERP systems, traceability platforms) are backward-looking, transaction-recording systems. They cannot encode complex, multi-jurisdictional regulatory logic, nor can they prevent non-compliant transactions before they are recorded. The radically new technology envisioned here is a multi-dimensional knowledge graph with embedded regulatory science, converging three previously unintegrated fields:

- Graph Theory & Event Sourcing: a Single Source of Truth (SSOT) encoding entities, relationships, and value flows as immutable, time-series events.
- Regulatory Science as Code: machine-readable encoding of jurisdictional regulations directly into graph schema constraints, enabling real-time compliance validation rather than post-hoc auditing.
- Sovereign AI Onboarding (BUD Agent): a natural-language-to-graph pipeline enabling non-technical actors (including smallholder farmers) to verify onto the network via conversational AI, reducing onboarding friction from weeks to minutes.

This convergence creates a Compliance Engine that is proactive rather than reactive, shifting the paradigm from 'trust but verify' to 'verify by design.' The INOS/SSOT/HEMPIN project (48 months, TRL 2–4) aims to validate the scientific and technological basis of this architecture, de-risking it for subsequent Innovation Actions and commercial deployment by 2030.

The vision aligns directly with EU strategic priorities: Strategic Autonomy (sovereign EU tech stack), the Green Deal (embedded regenerative funding), and the Digital Decade (AI adoption in SMEs). It positions Europe as the leader in verified supply chain infrastructure.

1.2 Science-towards-technology breakthrough

Core Breakthrough: Convergence of Graph Theory, Regulatory Science, and Sovereign AI

The project proposes a concrete science-towards-technology breakthrough: the development of a Regulatory-Encoding Knowledge Graph powered by a Sovereign AI Onboarding Pipeline. This is not an incremental improvement over existing supply chain technologies, but a fundamentally new architecture converging three scientific domains not previously integrated at this scale.

Breakthrough 1 - BUD Agent: Natural-Language-to-Graph (NL2G) Pipeline

State-of-the-Art: Current onboarding requires manual form-filling and weeks of compliance officer verification. AI solutions exist but rely on US-based APIs (OpenAI, AWS) and produce unstructured text outputs that cannot be verified or written directly to a graph.

INOS/SSOT Innovation: The BUD Agent uses a fine-tuned Mistral 7B model to convert natural language conversations (e.g., a farmer describing a harvest) directly into structured, verified graph nodes, including entity type, relationship, attribute values, and compliance status (e.g., THC < 0.3%).

Novelty: Integration of confidence scoring (threshold > 0.85) with human-in-the-loop validation for low-confidence extractions. Moves AI from chatbot (TRL 6 in generic chat) to verified data ingestion pipeline (TRL 4). Requires original research into hallucination mitigation for regulatory-critical data (WP3).

Breakthrough 2 - Regulatory Ontology as Graph Constraints

State-of-the-Art: Regulatory compliance is encoded as static PDFs or manual checklists. Blockchain solutions record transactions but cannot encode complex regulatory logic (e.g., THC limits that vary by jurisdiction, product type, and end-use). Verification is post-transaction.

INOS/SSOT Innovation: Regulatory rules are encoded as graph schema constraints (SHACL shapes, Cypher queries). A transaction cannot be recorded if it violates a constraint. The graph itself enforces compliance, eliminating the need for external auditing.

Novelty: Active prevention vs. passive detection. Requires original research into ontology mapping across multiple jurisdictions, resolving conflicts between incompatible regulatory frameworks (WP4).

Breakthrough 3 - Sovereign AI Stack

State-of-the-Art: Most AI supply chain tools rely on US-based cloud providers (AWS, Azure, Google) and APIs (OpenAI), exposing EU SMEs to extraterritorial laws (US Cloud Act) and creating data sovereignty risks.

INOS/SSOT Innovation: Full-stack EU sovereignty: OVHcloud (France) for hosting, Memgraph (Slovenia) for graph database, Mistral AI (France) for LLM inference. All infrastructure and sensitive commercial data remain within EU jurisdiction. Requires GPU infrastructure for self-hosted LLMs (justifies €200,000 equipment cost).

Novelty: Demonstrates that a sovereign EU stack can handle latency and throughput requirements for real-time AI inference; moving from theoretical feasibility (TRL 2) to validated performance benchmark (TRL 4) (WP2).

Comparison with State-of-the-Art

Feature	Current State-of-the-Art	INOS/SSOT Proposal	Advancement
Onboarding	Manual forms, email verification (weeks)	Conversational AI to Graph node (minutes)	Radical (10× speedup)
Compliance	Post-transaction auditing (PDFs)	Real-time graph constraint validation	Radical (Prevention vs. Detection)
AI Infrastructure	US-based APIs (OpenAI, AWS)	EU self-hosted Mistral on OVHcloud	Significant (Sovereignty)
Data Structure	Siloed databases, blockchains	Multi-dimensional SSOT knowledge graph	Significant (Interoperability)
Verification	Trust-but-verify (manual)	Verify-by-design (automated)	Radical (Paradigm shift)

TRL Progression

- Month 1 (Start): TRL 2 - Technology concept formulated.
- Month 24 (Mid-point): TRL 3 - Experimental proof-of-concept, lab validation of AI pipeline and 5-jurisdiction ontology.
- Month 48 (End): TRL 4 -Technology validated in lab/controlled environment with pilot data from Thailand and EU test environments.

This progression de-risks the technology for subsequent Horizon Europe Innovation Actions (EIC Accelerator, Clusters 4 and 6) and private investment, enabling full commercial deployment (TRL 6–7) by 2030.

1.3 Objectives

The project has 7 concrete, measurable, and verifiable objectives (O1–O7) aligned to Work Packages WP1–WP6, designed to achieve proof-of-principle (TRL 4) by Month 48.

Objective ID	Description	Related WP	Success Metric
O1	Establish project governance, ethical compliance framework, and consortium management.	WP1	Project Handbook approved Month 3; 100% reporting on time.
O2	Design and validate SSOT graph schema for hemp entities, relationships, and value flows.	WP2	Schema v1.0 published Month 12; supports 50+ entity types.
O3	Benchmark sovereign AI infrastructure (OVHcloud, Memgraph, Mistral) for real-time inference.	WP2	Latency <500 ms; 99.9% uptime benchmark validated.
O4	Build BUD Agent NL2G pipeline with confidence scoring >0.85 for entity onboarding.	WP3	85% auto-approval rate; hallucination rate <5% on test set.
O5	Encode regulatory rules for 5 jurisdictions as machine-readable graph constraints.	WP4	5 jurisdiction ontologies validated; non-compliant transactions blocked in test environment.
O6	Validate pilot model with 500+ entities onboarded in Thailand and EU environments.	WP5	500 verified entities; 50+ transactions processed; user satisfaction >4/5.
O7	Disseminate via 3+ open-access publications, open-source schema, and workshops.	WP6	3 publications (Q1/Q2 journals); GitHub repo with 100+ stars; 2 workshops held.

High-Risk/High-Gain Appropriateness

The project is inherently high-risk/high-gain, appropriate for EIC Pathfinder Open. The scientific risks are substantial: AI hallucination in regulatory contexts could lead to erroneous compliance decisions; encoding multi-jurisdictional regulatory logic as formal graph constraints may expose fundamental incompatibilities between legal frameworks; sovereign EU infrastructure may prove insufficient for real-time AI inference workloads at the required latency.

These are genuine research uncertainties, not engineering optimisation problems. Each requires original scientific investigation. The corresponding gains are transformative: successful validation would shift compliance from a cost centre to an infrastructure layer, reduce transaction costs by an estimated 30–40% for SMEs in fragmented regulated markets, and prove the feasibility of EU sovereign AI infrastructure for sensitive commercial data.

Overall Methodology

The project uses an agile research methodology combining iterative development with scientific validation through 6-month sprints and milestone-gated progression.

Core concept: A Regulatory-Encoding Knowledge Graph assumes that regulatory rules can be represented as graph schema constraints (SHACL, Cypher) without losing legal nuance or flexibility for local variations.

Models: Event-sourcing models for immutable transaction history; fine-tuned LLMs (Mistral 7B) for natural language processing; SHACL shapes for constraint validation.

Key assumptions: (1) Regulatory text can be structured without losing legal validity; validated in WP4 with legal experts. (2) Farmers prefer conversational onboarding over form-based systems; validated in WP5 pilot co-design. (3) Sovereign EU stack can meet <500 ms latency requirements; validated in WP2 benchmarking.

Alternative directions: If graph constraints prove too rigid for certain regulatory contexts, a hybrid validation layer (graph + external rules engine) will be implemented. If LLM fine-tuning yields insufficient accuracy, Retrieval-Augmented Generation (RAG) with stricter prompt engineering will be adopted as a fallback.

Do No Significant Harm (DNSH) Compliance

The methodology complies with Article 17 of Regulation (EU) No 2020/852. The project promotes digital efficiency, reducing travel for compliance audits. The Regeneration Fund protocol supports verified soil restoration and carbon sequestration projects. Hemp cultivation promoted via verified supply chains is a bio-based, low-input crop supporting circular economy objectives. The project does not involve hazardous chemicals, heavy manufacturing, or activities harming any of the six EU Taxonomy environmental objectives.

AI Robustness

The BUD Agent is an AI system that will have significant impact on supply chain compliance decisions. The following robustness measures are embedded in the methodology:

- Technical robustness: confidence scoring with > 0.85 threshold enforced before any graph write; fallback to manual verification for low-confidence extractions; regular accuracy audits at each reporting period.
- Social robustness: training data diversified across languages (EN, FR, TH, RO) and across farm typologies to reduce demographic bias; human-in-the-loop validation required for all regulatory-critical extractions.
- Reliability: 99.9% uptime SLA; error logging and incident response protocols documented in Project Handbook.
- Explainability: every graph node includes a provenance trail (source text, confidence score, human reviewer ID) enabling full auditability of AI decisions.

Gender Dimension in Research Content

The research content integrates sex and gender analysis in the following ways, recognising that agricultural supply chain digitisation has documented gender-differentiated outcomes:

- AI training data for the BUD Agent will be audited to ensure farmer personas used in training corpora are not male-defaulted. Hemp farming involves significant female participation, particularly in smallholder contexts in Southeast Asia; training data will explicitly include female farmer scenarios to prevent systematic bias in entity recognition.
- Pilot studies (WP5) will stratify usability feedback by gender to test whether the conversational interface performs equivalently for all users, with corrective fine-tuning if disparities are detected.
- The regulatory ontology (WP4) will examine whether encoded rules have gender-differential impacts (e.g., licensing requirements that disproportionately exclude smallholder women farmers) and flag these for policy briefings.

These measures address the research content, not team composition, in line with EIC requirements.

Open Science Practices

- Early and open sharing: key schema designs published as preprints (Zenodo, arXiv) before peer review. GitHub repository launched at Month 6 (Deliverable 6.1).
- Open access publications: all 3+ peer-reviewed papers published in Q1/Q2 open-access journals (Article Processing Charges budgeted in WP6).
- Open source code: SSOT graph schema and BUD Agent core pipeline released under MIT licence, enabling community contributions and preventing vendor lock-in.
- Reproducibility: complete test sets, model weights, and evaluation scripts released alongside publications.
- Citizen science: farmer pilots (WP5) involve end-users in co-creation of the BUD Agent interface; design decisions documented and published.

Research Data Management (FAIR Principles)

Types of data: Graph schema definitions, AI training datasets (anonymised), pilot transaction logs, regulatory ontologies. Estimated total size: 500 GB.

Findability: Persistent DOIs assigned via Zenodo; metadata indexed on GitHub and OpenAIRE.

Accessibility: Open access for schema, code, and publications; restricted access for sensitive pilot data (GDPR-compliant, available for verification on request to corresponding partner).

Interoperability: RDF, SHACL, and JSON-LD for graph/ontology data; CSV/Parquet for tabular experimental data; aligned to W3C and ISO standards.

Reusability: MIT licence for code and schema; CC-BY-4.0 for publications; CC-BY-NC for raw datasets.

Curation: INOS Technical Lead responsible for data quality and backup (OVHcloud, EU sovereignty). Data Management Plan delivered as Deliverable 1.2 (Month 6), revised at Month 42.

1.4 Interdisciplinarity

The problem addressed (enabling efficient, verified cross-border trade in highly fragmented regulatory environments) cannot be solved by any single discipline. Computer scientists build databases without encoding regulatory logic; legal scholars write regulations without considering machine-readability; agronomists optimise cultivation without digital verification layers. This siloed approach perpetuates the coordination failure that INOS/SSOT/HEMPIN aims to resolve.

The breakthrough (Regulatory-Encoding Knowledge Graphs) exists specifically at the intersection of five disciplines that do not routinely collaborate at this depth:

- Computer Science (Graph Theory & Event Sourcing): architectural foundation for immutable, time-series data modelling (WP2).
- Artificial Intelligence (NLP & LLMs): natural-language-to-graph conversion for low-friction onboarding (WP3).
- Regulatory Science & Law: translation of legal text across 57 jurisdictions into machine-readable graph constraints (WP4).
- Agricultural Science (Agronomy): validation that the graph schema reflects real-world hemp cultivation practices and biological constraints (WP5).
- Ecological Economics: design of the Regeneration Fund protocol, embedding environmental repair into the transaction layer (WP6).

New Scientific Collaborations

AI Engineers + Regulatory Lawyers (WP3/WP4): Lawyers define ontological constraints; engineers encode them as graph schemas (SHACL/Cypher). This treats regulation as a software specification, not a legal document, a genuinely novel collaboration modality.

Graph Database Architects + Agronomists (WP2/WP5): Agronomists validate the graph schema against real-world hemp cultivation data (harvest cycles, THC testing protocols, moisture content standards), ensuring the SSOT reflects biological reality.

Economists + Software Developers (WP2/WP6): Economists design the Regeneration Fund protocol (30% fee allocation logic); developers implement it as a deterministic graph-layer function. This embeds economic policy directly into infrastructure.

Integration in the Work Plan

The interdisciplinary approach is structurally enforced through the work plan, not merely co-located:

- WP2 (Architecture) and WP4 (Compliance) are iterative: graph schema design cannot be finalised without regulatory constraints; regulatory constraints must be tested against the schema.
- WP3 (AI) and WP5 (Pilots) are interdependent: BUD Agent training (AI) uses real farmer conversation data (Agronomy); pilot feedback drives model re-fine-tuning.
- WP6 (Dissemination) produces joint publications spanning disciplines (e.g., 'AI for Regulatory Compliance', 'Graph Databases for Agronomic Traceability'), requiring active cross-disciplinary co-authorship.

The coordinator (INOS) brings core expertise in computer science and AI. Prospective consortium partners (including a leading EU hemp industry association (regulatory expertise, WP4) and a European agricultural research university (agronomy, WP5)) provide domain knowledge not available within INOS. Subcontractors (ML engineering, legal analysis) bridge remaining gaps. This ensures the project has access to all five disciplines without over-extending any single partner.

2. Impact

2.1 Long-term impact

The INOS/SSOT/HEMPIN project's direct outcome is a validated proof-of-principle (TRL 4) for Regulatory-Encoding Knowledge Graphs in the hemp industry. Its transformative effects, however, extend far beyond the 48-month project duration across three dimensions.

(a) Transformative Positive Effects

Economy: Unlocking Efficient Cross-Border Trade for SMEs

The hemp industry ($17B global market, 25% CAGR [EIHA Market Report 2024]) currently operates as a coordination failure: 57 jurisdictions with incompatible regulations, no shared identity standard, and no verified trust layer. This fragmentation imposes prohibitive transaction costs on SMEs — farmers, processors, and brands — who must manually verify compliance for each cross-border trade. Customs rejections due to non-compliance cost the EU hemp sector an estimated €250M annually.

When fully deployed (2030+), the INOS/SSOT technology is projected to reduce transaction costs by 30–40% for SMEs by automating compliance validation. It unlocks global markets for European hemp SMEs currently blocked by regulatory complexity (e.g., French hemp brands accessing certified Asian markets). It creates a Verification-as-a-Service (VaaS) market that generalises to organic agriculture, textiles, and critical minerals — any supply chain requiring verified trust across jurisdictions.

Environment: Regenerative Economics Embedded in Infrastructure

The project's architecture includes a Regeneration Fund protocol: 30% of transaction fees are automatically routed to verified environmental cleanup projects (soil remediation, water restoration, carbon sequestration). This is not a charitable add-on; it is encoded into the graph's value layer as a deterministic function.

Long-term (2030–2045), assuming 1% market penetration in verified supply chains, this protocol could route over €10B cumulatively to verified environmental repair — a sustainable funding mechanism independent of philanthropy or government grants, operationalising the 'polluter pays' principle in reverse.

Society: Data Sovereignty and Democratic Access

By building on a sovereign EU technology stack, the infrastructure ensures sensitive commercial data never leaves EU jurisdiction, reducing European SME exposure to extraterritorial laws (US Cloud Act). The BUD Agent's conversational interface lowers technical barriers, enabling smallholder farmers — not only large corporations — to participate in verified supply chains. The graph encodes compliance without exposing commercial relationships, balancing verification with privacy.

Potential Negative Environmental Impact

The project acknowledges that self-hosted LLM inference at commercial scale carries a significant energy footprint. At project scale (research infrastructure), this is managed by using OVHcloud green data centres (renewable energy certified) and model quantisation to reduce compute per inference. Residual energy consumption is offset through the Regeneration Fund protocol. At commercial scale, the project commits to annual carbon footprint reporting and to publishing energy efficiency benchmarks alongside performance results, enabling the research community to evaluate the trade-offs honestly.

(b) Requirements, Barriers, and Mitigating Measures

The transformative effects described above depend on factors beyond the project's scope and duration. The table below identifies the most significant barriers and proposed mitigating measures.

Barriers:

Barrier	Description	Likelihood	Severity
Regulatory Fragmentation	New jurisdictions add complexity faster than ontology can encode them.	Medium	High
Adoption Friction	Farmers prefer existing workflows (Excel, WhatsApp) over digital onboarding.	Medium	Medium
Sovereign Stack Performance	EU infrastructure may not match US API latency at scale.	Low	High
Competing Platforms	US-based traceability platforms may capture market share first.	Medium	Medium
Post-Grant Funding Gap	Transition to commercial deployment requires private investment.	Medium	High
GPU Energy Footprint	Self-hosted LLM inference at scale has significant energy consumption; potential negative environmental impact when brought to commercial scale.	Low	Medium

Mitigating Measures:

Barrier	Mitigation Within Project	Mitigation Beyond Project
Regulatory Fragmentation	Encode 5 jurisdictions initially (WP4); design ontology for extensibility; EIHA/InterChanvre partners provide regulatory monitoring.	Establish Regulatory Ontology Working Group post-project; open-source schema enables community contributions.
Adoption Friction	BUD Agent uses conversational AI; pilot incentives via association partners; human-in-the-loop console.	Freemium model for smallholders; association partnerships drive network effects; pilot success stories create social proof.
Sovereign Stack Performance	Benchmark EU stack vs. US alternatives (WP2); GPU optimization (€200K); modular architecture allows hybrid deployment.	Partner with EU cloud providers for dedicated infrastructure; leverage EU strategic autonomy policies.
Competing Platforms	Differentiate via regulatory encoding; sovereign stack appeals to EU SMEs concerned about data sovereignty.	Open-source core schema to prevent vendor lock-in; focus on hemp niche first.
Post-Grant Funding Gap	Design for commercial viability from start (VaaS model); engage prospective investors via advisor board.	Pursue EIC Accelerator follow-on; attract impact investors; revenue from VaaS subscriptions.
GPU Energy Footprint	Optimise inference efficiency (model quantisation, batching); use OVHcloud green data centres (renewable energy certified).	Offset residual energy consumption via the Regeneration Fund protocol; publish annual carbon footprint reports.

Direct Contribution to EIC Pathfinder Expected Impacts

This project makes a significant and direct contribution to the EIC Pathfinder Open expected impacts. The scientific contribution validates the convergence of graph theory, regulatory science, and sovereign AI — a novel interdisciplinary breakthrough. The technological contribution delivers a TRL 4 proof-of-principle for a radically new compliance architecture. The economic contribution demonstrates a pathway to reduce SME transaction costs in fragmented markets. The environmental contribution embeds regenerative economics into the infrastructure layer. The societal contribution advances EU data sovereignty, aligning with the EU's Strategic Autonomy agenda.

2.2 Innovation potential

Disruptive Innovation: Verification-as-a-Service (VaaS)

The INOS/SSOT/HEMPIN project has high potential for generating disruptive innovation and creating new markets beyond the project duration. The core innovation — Regulatory-Encoding Knowledge Graphs — enables a new market category: Verification-as-a-Service (VaaS). Unlike existing supply chain technologies (blockchain traceability, ERP systems, compliance SaaS), VaaS does not merely record or audit transactions. It actively prevents non-compliant transactions from being recorded by encoding regulatory rules as machine-readable constraints. This paradigm shift — from 'trust but verify' to 'verify by design' — creates a new market for proactive compliance infrastructure.

Market potential (conservative estimates): Hemp industry (Phase 1) — $17B market, 1–2% VaaS capture by 2030 (~€170–340M annual revenue). Regulated agriculture (Phase 2, 2030–2035) — $200B+ market, 0.5–1% capture (~€1–2B annual revenue). Global verified trade infrastructure (Phase 3, 2035–2045) — multi-trillion dollar addressable market.

Intellectual Property Strategy

IP Type	What Is Protected	Protection Mechanism	Rationale
Core Algorithm	NL2G pipeline with confidence scoring	EU + PCT patent filing (Month 24)	Prevents replication by US-based AI platforms; enables licensing revenue.
Graph Schema	SSOT entity/relationship/compliance/value layer definitions	Open source — MIT licence	Encourages adoption and community contributions; creates network effects.
Regulatory Ontology	Machine-readable encoding of jurisdictional rules	Copyright + open access (CC-BY-4.0)	Transparency while preventing proprietary lock-in.
Sovereign Stack Architecture	EU deployment pattern (OVHcloud + Memgraph + Mistral)	Trade secret + technical documentation	Differentiates from US-dependent alternatives; supports EU strategic autonomy.

This hybrid IP strategy balances openness (to drive adoption and network effects) with protection (to enable commercialisation). Results ownership will be specified in the Consortium Agreement and the final periodic report results ownership list, as required by Horizon Europe.

Exploitation Beyond the EU

The Thailand pilot (TiHTA LOI) implies exploitation activity in a non-associated third country. This is justified as being in the EU interest for the following reasons: (1) the Thailand pilot provides high-diversity regulatory and agronomic validation data not available in EU markets alone, strengthening the scientific robustness of the proof-of-principle; (2) demonstrating the SSOT system in a non-EU regulatory context proves the architecture's generalisability, making the resulting technology more attractive to EU and global markets; (3) any commercial exploitation of results will be led by INOS (France), an EU-established entity, with intellectual property ownership remaining within the EU consortium. No EU funding is requested for TiHTA, which participates as an unfunded associated partner.

Organisational Structure for Exploitation

The project establishes a Foundation + OpCo structure to manage exploitation post-grant: the INOS Foundation (non-profit) holds the open-source schema, regulatory ontology, and Regeneration Fund protocol, ensuring mission

lock; the INOS OpCo (for-profit SASU) licenses the patented NL2G algorithm and provides VaaS subscriptions. This structure is finalised during grant agreement preparation (Months 1–6).

Involving Key Actors for Future Innovation

- Early-career researchers: INOS commits to hiring 2 PhD students (WP2/WP3) with clear pathways to post-doctoral or startup roles; research outputs co-authored to build publication records.
- High-tech SMEs and startups: the open-source schema enables SMEs to build complementary tools; INOS will host hackathons (WP6) to encourage SME participation; startups may license the NL2G algorithm for niche applications.
- Farmer cooperatives: pilot design (WP5) ensures smallholder farmers co-create the BUD Agent interface, empowering them to access global markets through the verified network.
- Standardisation: INOS will contribute the SSOT schema to ISO/TC 307 (blockchain and distributed ledger technologies) and CEN/TC 444 (hemp products) to promote interoperability and support future regulatory harmonisation.

2.3 Communication and dissemination

The project implements a multi-layered dissemination and communication strategy targeting four primary audiences: the scientific community, hemp industry SMEs and end users, financial actors, and the public at large. All activities are strategically planned from project outset and proportionate to the project scale.

Target Group	Key Messages	Channels & Activities	Timing
Scientific Community	Novel NL2G pipeline; FAIR data for supply chain; EU sovereign AI stack.	3+ open-access publications; preprints on Zenodo; conferences (ESWC, ISWC).	Months 18, 36, 48
Hemp Industry SMEs	Verified supply chains reduce compliance friction; BUD Agent lowers onboarding barriers.	Workshops at Cannatec, EIHA Annual Conference; webinars; LinkedIn weekly series.	Months 6–48
Financial Actors	VaaS market potential €170M–2B by 2035; Foundation+OpCo structure.	EIC Accelerator pitch prep; investor briefings via advisor network; impact reports.	Months 36–48+
Public at Large	Verified supply chains enable sustainable consumption; data sovereignty empowers SMEs.	Landing page with explainer videos (EN/FR/TH); social media campaigns; press releases.	Months 1–48

Concrete Actions by Year

Year 1 (Months 1–12): Launch landing page with plain-language vision explainer (Month 1); publish preprint of SSOT graph schema design on Zenodo (Month 6); launch open-source GitHub repository (Month 6); present at 1 major hemp industry event (Month 9); release first open-source component (Month 12).

Year 2 (Months 13–30): Publish first peer-reviewed paper (Month 18); host 2 webinars demonstrating BUD Agent onboarding flow (Months 20, 24); release Thailand pilot results as open FAIR dataset (Month 24); present at 1 academic and 1 industry conference.

Years 3–4 (Months 31–48): Publish second and third peer-reviewed papers (Months 36, 48); release full SSOT schema v1.0 (Month 36); host final project workshop (Month 42); submit Dissemination and Exploitation Plan update; publish post-project exploitation summary.

Policy Feedback

The project generates policy-relevant outputs for EU DG AGRI, DG CONNECT, and DG CLIMA: a replicable methodology for machine-readable regulation (Digital Europe Programme), sovereign stack performance benchmarks supporting EU strategic autonomy policy, and the Regeneration Fund model as a template for the Green Deal's circular economy action plan. Policy briefs (2–4 pages) will be distributed via prospective association partners who have established policy engagement channels.

A detailed Dissemination and Exploitation Plan including communication activities will be delivered as Deliverable 4.2 by Month 6 and updated at each reporting period, as required by Horizon Europe.

AI Tool Disclosure

The preparation of this proposal involved the use of AI-assisted drafting tools. All content has been thoroughly reviewed, validated, and corrected by the applicants. Sources cited have been verified independently. The applicants take full responsibility for the accuracy and appropriateness of the proposal content, including all AI-assisted sections.

3. Quality and efficiency of the implementation

3.1 Work plan and allocation of resources

Overall Structure

The project is structured into 6 Work Packages (WP1–WP6) over 48 months, following an agile research methodology with 6-month iterative development cycles and milestone-gated progression. The work plan follows a clear logical dependency chain: WP2 (architecture) provides the schema foundation for WP3 (AI) and WP4 (compliance), which in turn feed WP5 (pilots). WP6 (dissemination) and WP1 (management) run throughout.

Reporting Periods

Project Duration	RP1 Duration	RP2 Duration	RP3 Duration	Total Months
48 months	Months 1–12	Months 13–30	Months 31–48	48

Gantt Chart — Timeline Overview

The table below provides a colour-coded Gantt representation (■ = active period) across 8 bi-monthly blocks.

Work Package	M1–6	M7–12	M13–18	M19–24	M25–30	M31–36	M37–42	M43–48
WP1 – Management	■	■	■	■	■	■	■	■
WP2 – Graph Architecture	■	■	■	■	■	■		
WP3 – BUD AI Agent		■	■	■	■	■	■	
WP4 – Compliance Engine		■	■	■	■	■	■	■
WP5 – Pilot Deployment			■	■	■	■	■	■
WP6 – Dissemination	■	■	■	■	■	■	■	■

Component Inter-relations (PERT Summary)

WP2 is the critical path dependency: graph schema (D2.1, Month 12) must be finalised before WP3 can complete BUD Agent integration and before WP4 can validate constraint encoding. M2 (Schema Finalised, Month 12) gates the start of full WP3 and WP4 activity. M4 (Compliance Engine Live, Month 24) gates WP5 pilot launch. M8 (TRL 4 Validated, Month 42) is the primary scientific milestone, leading to final reporting and exploitation planning (M9, Month 48).

Table 3.1a — List of Work Packages

WP No.	WP Title	Lead P. No.	Short Name	WP Leader Name	Gender	Start	End
WP1	Project Management & Coordination	1	INOS	Paul IGLESIA	M	1	48
WP2	SSOT Graph Architecture & Sovereign Infrastructure	1	INOS	[Technical Lead – TBD at GA prep]	[TBD]	1	36
WP3	BUD AI Agent Development (NL2G Pipeline)	1	INOS	[AI Lead – TBD at GA prep]	[TBD]	6	42
WP4	Compliance Engine & Regulatory Encoding	2	[Partner 2]	[Regulatory Lead – TBD]	[TBD]	6	48
WP5	Pilot Deployment & Validation	3	[Partner 3]	[Pilot Lead – TBD]	[TBD]	12	48
WP6	Dissemination, Ethics & Open	1	INOS	[Comms Lead –	[TBD]	1	48

WP No.	WP Title	Lead P. No.	Short Name	WP Leader Name	Gender	Start	End
	Science			TBD at GA prep]			

Table 3.1b — Work Package Descriptions

WP1: Project Management & Coordination (Months 1–48, 16 PM)

Objectives: Ensure governance, ethical compliance, GDPR compliance, and timely reporting.

Tasks: (1.1) Consortium coordination — monthly calls, progress reporting, escalation procedures; 6 consortium calls per year, 3 periodic reports. (1.2) Ethical and legal compliance — GDPR data processing agreements with all partners; AI Act alignment review (Month 12). (1.3) Risk management — quarterly risk register updates (Table 3.1e); corrective action protocol. (1.4) Financial management — budget tracking against WP allocation; subcontracting oversight. Lead: INOS (Paul Iglesia). Total effort: 16 PM (INOS 12 PM, Partners 4 PM).

WP2: SSOT Graph Architecture & Sovereign Infrastructure (Months 1–36, 40 PM)

Objectives: Design SSOT graph schema; deploy and benchmark sovereign EU AI stack.

Tasks: (2.1) SSOT schema design — define 4 graph layers (Entity, Relationship, Compliance, Value); 50+ entity type specifications; 3 schema review cycles with WP4. (2.2) Sovereign stack deployment — OVHcloud hosting setup; Memgraph deployment and configuration; Mistral 7B model deployment on GPU servers; 2 load tests (Month 9, Month 24). (2.3) Performance benchmarking — compare EU stack vs. US API alternatives on 6 metrics (latency, throughput, cost, availability, energy, compliance); benchmark report at Month 18 (D2.2). (2.4) Security and sovereignty audit — third-party GDPR/security audit (Month 36, M7); publication of audit findings. Lead: INOS. Effort: 40 PM (INOS).

WP3: BUD AI Agent Development (Months 6–42, 40 PM)

Objectives: Build NL2G pipeline with confidence scoring > 0.85 for entity onboarding.

Tasks: (3.1) Ontology refinement for AI training — prepare 3,000+ annotated conversational examples across EN, FR, TH, RO; gender-balanced farmer persona corpus. (3.2) LLM fine-tuning — fine-tune Mistral 7B on annotated corpus; 4 training iterations with evaluation (Months 9, 15, 21, 27). (3.3) Confidence scoring implementation — design and validate threshold classifier; target hallucination rate < 5% on regulatory test set; 2 independent evaluations. (3.4) Human-in-loop console development — web console for compliance officer review of low-confidence extractions; UX testing with 20 compliance officers (Month 30). Lead: INOS. Effort: 40 PM (INOS).

WP4: Compliance Engine & Regulatory Encoding (Months 6–48, 20 PM)

Objectives: Encode regulatory rules for 5 jurisdictions as machine-readable graph constraints.

Tasks: (4.1) Regulatory ontology mapping — legal analysis of hemp regulations in TH, FR, DE, NL, AU; identification of 100+ regulatory rules per jurisdiction; conflict resolution protocol for overlapping rules; led by Partner 2 (regulatory specialist). (4.2) Graph constraint encoding — convert regulatory rules to SHACL shapes and Cypher constraints; 5 jurisdiction ontologies validated against 200+ test transactions each. (4.3) Compliance engine validation — automated test suite with > 200 compliant and > 200 non-compliant transaction scenarios; false positive rate < 2%. (4.4) Iterative expansion plan — document extensibility methodology for additional jurisdictions post-grant. Lead: Partner 2. Effort: 20 PM (INOS 4 PM, Partner 2 16 PM).

WP5: Pilot Deployment & Validation (Months 12–48, 22 PM)

Objectives: Validate the integrated system with 500+ entities in Thailand and EU test environments.

Tasks: (5.1) Thailand pilot setup — TiHTA partnership operationalisation; ethical approval for data collection; onboarding of 250+ Thai hemp entities via BUD Agent (Months 18–30). (5.2) EU pilot setup — onboarding of

250+ EU hemp entities (farmers, processors, brands) via association partner networks; GDPR-compliant data processing (Months 24–36). (5.3) Transaction validation — process 50+ cross-border transactions through Compliance Engine; record compliance outcomes; measure time-to-compliance vs. baseline manual process. (5.4) Feedback and iteration — user satisfaction surveys (target > 4/5); stratified by gender; feed results back to WP3 for model improvement. Lead: Partner 3. Effort: 22 PM (INOS 4 PM, Partner 2 4 PM, Partner 3 14 PM).

WP6: Dissemination, Ethics & Open Science (Months 1–48, 9 PM)

Objectives: 3+ open-access publications; open-source schema release; stakeholder engagement; ethics compliance.

Tasks: (6.1) Publications — draft and submit 3 peer-reviewed papers (Months 18, 36, 48); preprints published 3 months prior to submission. (6.2) Open-source repository — GitHub launch (Month 6); schema v1.0 release (Month 36); community contribution guidelines; 4 documentation updates. (6.3) Stakeholder events — 2 industry workshops (Cannatec, EIHA Annual Conference); 2 webinars; 1 final project symposium (Month 42). (6.4) Policy engagement — 2 policy briefs (Months 24, 48); distributed via Partner 2 (association) policy channels. (6.5) Ethics monitoring — quarterly ethics checklist; AI Act alignment updates as legislation develops. Lead: INOS. Effort: 9 PM (INOS 5 PM, Partners 4 PM).

Table 3.1c — List of Deliverables

No.	Deliverable Name	Short Description	WP	Lead	Type	Dissem.	Month
1.1	Project Handbook	Management procedures, comm. plan	1	INOS	R	PU	3
1.2	Data Management Plan (DMP)	FAIR data management strategy	1	INOS	DMP	PU	6
2.1	SSOT Graph Schema v1.0	Entity/Relationship/Compliance/Value layer specs	2	INOS	R	PU	12
2.2	Sovereign Stack Benchmark Report	Performance metrics vs. US alternatives (latency, throughput, energy)	2	INOS	R	PU	18
3.1	BUD Pipeline v1.0	NL2G pipeline code + documentation	3	INOS	OTHER	PU	18
3.2	Confidence Scoring Model Report	Technical report on scoring methodology + hallucination audit	3	INOS	R	PU	24
4.1	Regulatory Ontology (5 jurisdictions)	Machine-readable rules for TH, FR, DE, NL, AU	4	P2	DATA	PU	24
4.2	Dissemination & Exploitation Plan	Full plan per HE requirements, first version	6	INOS	R	PU	6
5.1	Thailand Pilot Report	Results from TiHTA pilot (500 entities, anonymised)	5	P3	R	PU	30
5.2	EU Pilot Report	Results from EU test environment	5	P2	R	PU	36
6.1	Open Source Repository	GitHub repository with code, schema, documentation (MIT)	6	INOS	DEC	PU	6
6.2	Scientific Publications (3+)	Peer-reviewed papers on graph AI, compliance encoding, sovereignty	6	INOS	R	PU	48

Table 3.1d — List of Milestones

MS No.	Milestone Name	Related WP(s)	Due Month	Means of Verification
M1	Consortium Agreement Signed	WP1	1	Signed CA uploaded to portal.
M2	SSOT Graph Schema v1.0 Finalised	WP2	12	Schema document + internal peer review report.
M3	BUD Agent Alpha (Confidence ≥0.85)	WP3	18	Test report with accuracy metrics on held-out test set.
M4	Compliance Engine Live (5 Jurisdictions)	WP4	24	Working prototype + validation report confirming blocked non-compliant test transactions.
M5	Thailand Pilot Launch	WP5	24	Pilot launch report + participant list (≥50 entities registered).
M6	EU Pilot Launch	WP5	30	Pilot launch report + participant list (≥50 entities registered).
M7	Sovereign Stack Security & Performance Audit	WP2	36	Third-party audit report confirming GDPR compliance and latency benchmarks.
M8	Proof of Principle Achieved (TRL 4)	WP2,3,4,5	42	Technical validation report demonstrating end-to-end NL2G compliance validation on pilot data.
M9	Final Project Review & Exploitation Plan	WP1,6	48	Final report accepted + exploitation roadmap published.

Table 3.1e — Critical Risks for Implementation

Description of Risk (likelihood / severity)	WP(s)	Proposed Risk-Mitigation Measures
AI extraction accuracy <85% (Likelihood: Medium / Severity: High)	WP3	Human-in-the-loop console; fallback to manual verification; continuous fine-tuning on hemp ontology; confidence threshold enforced before any graph write.
Regulatory encoding too complex for graph constraints (Likelihood: High / Severity: Medium)	WP4	Start with 5 jurisdictions; design ontology for extensibility using modular SHACL shapes; Partner 2 (regulatory expert) provides legal validation; hybrid rules-engine fallback if pure graph constraints prove insufficient.
Sovereign stack performance issues at scale (Likelihood: Low / Severity: Critical)	WP2	Benchmark vs. US alternatives (M7); model quantisation to reduce GPU load; backup provider identified (Hetzner); modular architecture allows hybrid cloud/US-API fallback in non-sensitive contexts.
Pilot adoption friction — actors resist digital onboarding (Likelihood: Medium / Severity: Medium)	WP5	Conversational AI interface minimises friction; free tier for smallholders; incentives via association partners; iterative UX co-design with pilot users.
Graph database does not scale to pilot load (Likelihood: Low / Severity: High)	WP2	Memgraph load testing with synthetic pilot data (Month 6); sharding strategy designed in WP2; Memgraph vendor SLA secured.
Partner 2/3 (prospective) not confirmed in time (Likelihood: Medium / Severity: High)	WP1,4,5	Consortium formalised at grant agreement preparation; INOS subcontracting budget (€300K) covers regulatory and agronomy expertise if a partner drops out; TiHTA LOI provides pilot access.
GPU energy footprint raises environmental concerns (Likelihood: Low / Severity: Medium)	WP2	Use OVHcloud green data centres (renewable energy certified); model quantisation and inference batching reduce energy per query; annual carbon footprint reporting included in WP6.

Table 3.1f — Summary of Staff Effort (Person-Months)

Note: WP leaders shown in bold in individual WP descriptions above. Partner 2 = prospective regulatory specialist partner; Partner 3 = prospective agronomy/pilot partner. Full names and PM breakdowns to be confirmed at Grant Agreement preparation.

WP / Participant	1 – INOS (PM)	2 – Partner 2 (PM)	3 – Partner 3 (PM)	Total PM per WP
WP1 – Project Management	12	2	2	16
WP2 – Graph Architecture & Infra	40	–	–	40
WP3 – BUD AI Agent	40	–	–	40
WP4 – Compliance Engine	4	16	–	20
WP5 – Pilot Deployment	4	4	14	22
WP6 – Dissemination & Open Science	5	2	2	9
TOTAL PM per Participant	**105**	**24**	**18**	**147**

Table 3.1g — Subcontracting Costs

Participant No./Short Name	Cost (€)	Description of Tasks and Justification
1 / INOS	300,000	AI fine-tuning specialist (Mistral 7B optimisation and hallucination mitigation — WP3). Graph database optimisation consultant (Memgraph query performance — WP2). UI/UX designer for BUD conversational console (WP3). Core tasks — graph schema design, regulatory encoding, pilot management — are NOT subcontracted and remain with consortium beneficiaries.

Table 3.1h — Major Equipment Purchase Costs (>15% of personnel costs)

Participant No./Short Name	Cost (€)	Justification
1 / INOS	200,000	GPU servers for self-hosted Mistral 7B inference (sovereign AI stack, WP2/WP3). INOS personnel costs: €1,200,000; equipment ratio: 16.7% (exceeds 15% threshold, justification mandatory). Required for: (1) sovereign AI stack without US API dependency; (2) self-hosted LLM inference for BUD Agent; (3) EU data sovereignty compliance. OVHcloud renewable-energy certified data centres used to minimise energy footprint. Full justification also provided in the Detailed Budget Table annex (lump sum).

3.2 Quality of the consortium

Overview

The consortium is designed to cover all five interdisciplinary domains required by the project: computer science/AI (INOS), regulatory science (Partner 2), and agronomy/pilot operations (Partner 3). INOS serves as coordinator, leading the technical core (WP1, WP2, WP3, WP6). Partners 2 and 3 bring domain expertise not available within INOS, leading WP4 and WP5 respectively. Consortium finalisation is underway; confirmed LOIs will be provided at Grant Agreement preparation stage. INOS's subcontracting budget (€300,000) provides a fallback ensuring core task coverage if consortium composition changes.

INOS Foundation (France) — Coordinator

Core expertise in computer science, graph databases, AI pipeline development, and product management. Paul Iglesia (Founder, WP1 Lead) has prior experience as chef de projet logiciel, with validated technical competence in

graph database architecture, AI integration, and stakeholder coordination, evidenced by the Thailand LOI with TiHTA, 20+ files of technical documentation on GitHub, and the ADIE-validated business plan. INOS leads WP1 (Management), WP2 (Architecture), WP3 (AI), and WP6 (Dissemination), covering the full technical core. This ensures the scientific and technological objectives are achievable by the confirmed beneficiary independent of consortium finalisation.

Prospective Partner 2 — Regulatory Specialist (EU Member State or Associated Country)

Advanced discussions with a leading EU hemp industry association (and/or specialist regulatory law firm) providing regulatory science expertise for WP4. This partner brings: access to legal expertise across 57+ jurisdictions; established policy channels (EU DG AGRI, national competent authorities); networks for EU pilot recruitment. Role: WP4 Lead (Regulatory Encoding); co-lead of WP5 (EU Pilots); policy briefings (WP6). Effort: 24 PM.

Prospective Partner 3 — Agricultural Research Institution (EU Member State or Associated Country)

Advanced discussions with a European agricultural research university providing agronomy and pilot operations expertise for WP5. This partner brings: access to hemp farming networks in at least one EU Member State; established data collection protocols for agricultural trials; expertise in farmer-facing technology adoption studies. Role: WP5 Lead (Pilot Deployment); agronomic schema validation (WP2); user research (WP3). Effort: 18 PM.

Associated Partner: TiHTA (Thailand Hemp Industry Association) — Unfunded

TiHTA provides access to the Thailand pilot site (250+ hemp entities) under a signed Letter of Intent. No EU funding is requested. TiHTA's participation is essential because: it provides regulatory and agronomic diversity not available in EU markets alone, substantially strengthening the proof-of-principle's generalisability; it gives access to smallholder farming contexts representative of the global hemp supply chain; commercial exploitation of results arising from the Thailand pilot will be led by INOS (France), with IP remaining within the EU consortium.

Access to Critical Infrastructure

- Sovereign Cloud: OVHcloud (France) — hosting infrastructure; GDPR-compliant; renewable-energy certified data centres.
- Graph Database: Memgraph (Slovenia, EU-based) — high-performance in-memory graph storage for real-time compliance validation.
- AI Infrastructure: GPU servers (self-hosted, €200,000 equipment budget) for Mistral 7B inference — sovereign, without US API dependency.
- Pilot Sites: Thailand (TiHTA LOI, 250+ entities); EU (via Partner 2/3 networks, 250+ entities).

Complementarity and Interdisciplinary Coverage

The consortium's expertise map directly supports the interdisciplinary approach described in Section 1.4. INOS (CS/AI) + Partner 2 (Law/Regulatory Science) + Partner 3 (Agronomy) + subcontractors (ML engineering, UX design) covers all five required disciplines. Each partner contributes knowledge unavailable in the others, ensuring genuine complementarity rather than duplication.

Gender Balance and Diversity

INOS commits to gender-balanced hiring for technical roles (target 40% women in WP2/WP3 teams). Prospective partners are selected with attention to gender diversity in leadership roles. Gender analysis is integrated into the research content as described in Section 1.3. The consortium spans at minimum 3 EU Member States or Associated Countries upon finalisation, satisfying the EIC Pathfinder multi-country consortium requirement.